Unit 2 – Selection and Iteration Name: **KEY**
HW 5

1) Given the following code, fill in the blank with the correct information to display the desired output.

Teacher Note: There are a few correct ways this can be done!

a)
```
int count =  1;
while(count <= 8)
{
      System.out.print(count + " ");
      count = count + 1;
}
```

Output: 1 2 3 4 5 6 7 8

b)
```
int count =  1;
while(count < 12)
{
      System.out.print(count + " ");
      count = count + 2;
}
```

Output: 1 3 5 7 9 11

c)
```
int count =  1;
while(count < 15)
{
      count = count + 3;
      System.out.print(count + " ");

}
```

Output: 4 7 10 13 16

d)
```
int count =  1;
while(count < 13)
{
      count = count + 2;
      System.out.print(count + " ");
      count = count + 1;

}
```

Output: 3 6 9 12

2) The following algorithms attempts to print off the digits in a number in reverse order, but there is an error. Find and correct the error.

```
int number = 123456;
while(number > 0){
      System.out.print(number % 10);
      number = number - number % 10; number / 10
}
```

3) The following algorithm attempts to count the number of "E"s in a String, but there are TWO errors. Find and correct the errors.

```
String word = "BEEKEEPER";
int index = 0;
int countE = 0;
while(index < word.length()) word.length() - 1 {
      if(word.substring(index, index + 1).equals("E")){
            countE++;
            index++;
      } index++;
}
```

B 4) The following loop is intended to print the even numbers from 20 to 26 inclusive:

```
int x = 20;
while(x < 26)
{
  System.out.print(x);
  x++;
}
```

What would you need to change in order for the code to work correctly?

(A) It needs an if statement: `if (x % 2 == 0)`
(B) The `x++` needs to be `x += 2` and the `x < 26` needs to be <=
(C) The `x < 26` needs to be <=
(D) The `x++` needs to be `x += 2`

5) Complete the following method so that every odd number from start to finish is displayed, including start and finish if they are odd.

Example Code	Displays
`printOdd(3, 15)`	3 5 7 9 11 13
`printOdd(2, 10)`	3 5 7 9

```
public printOdd(int start, int finish) {

    if(start % 2 == 0) {
        start++;
    }
    if(finish % 2 == 0) {
        finish--;
    }

    while(start <= finish)
    {
        System.out.print(start + " ");
        start += 2;
    }




}
```